

Activity 9 Instruction Manual

Activity 9 | Topic 01 | C013 | Version v1.0

Purpose

You will turn your work packages into a sequenced, estimated schedule, draw the network diagram, and compute the critical path - then check whether GenAI's arithmetic can be trusted.

Files used

- activity-guide.md and activity-guide.pdf - brief and source instructions
- prompts.pdf - reusable prompt and verification checklist
- mock-data.xlsx - safe fictional project data for practice

Expected output

A Gantt chart, a schedule network diagram and a calculated critical path.

Before you begin

1. Form the required group and assign one person to capture evidence.
2. Open mock-data.xlsx and review the title, note and column headings.
3. Open prompts.pdf beside your chosen GenAI tool. Do not upload confidential workplace data.

Detailed procedure

1. In your group, prompt GenAI: 'Generate a list of tasks with estimated durations for building a CRM system over 8 weeks.' Open 'Gantt Chart Template.xlsx'.
2. Enter the tasks and durations, then add dependencies. Classify each dependency as Mandatory, Discretionary, External or Internal - GenAI will not do this correctly for your context.
3. Apply three-point estimating to your five most uncertain tasks. Prompt: 'For each task give an optimistic, most likely and pessimistic duration.' Compute the PERT estimate yourself: $(O + 4M + P) / 6$.
4. Compare the PERT figure with GenAI's single-point estimate. Note how much optimism was hidden in the original number.
5. Open 'Schedule Network Diagram.xlsx'. Prompt: 'List 10 project tasks with durations and dependencies for developing an internal helpdesk ticketing system.'
6. Enter activities, durations and dependencies, then perform the forward and backward pass to find each activity's float.
7. Determine the critical path - the sequence with zero float - and calculate the total project duration.
8. Now verify GenAI: ask it to compute the critical path for the same network, and compare with your manual answer. Where they differ, work out which is right and why.
9. Share your critical path and total duration with the class.

Use the supplied mock data

Copy only the rows needed for the current step and tell the model the data is fictional. If the model changes an identifier or number, reject the change and rerun with an explicit preserve-values instruction.

Quality control

- You can state the critical path, the total duration and the float on at least one non-critical activity, and you have checked GenAI's computed path against your own.
- Compare the first AI draft with the final human-corrected artifact and retain both as evidence.
- Label assumptions, decisions, owners and dates so another project manager can audit the result.

Troubleshooting

If the response is generic, add the project objective, constraints, audience and required columns. If arithmetic is inconsistent, request formulas and substitutions, then calculate independently. If a source cannot be opened, remove the claim or replace it with verified evidence.

Submission evidence

Submit the completed project artifact, the final prompt, a short critique of the AI output, and any screenshot or exported file required by the activity.